

Tanmay Mirgal

Full-Stack Engineer | ML/DL Developer | AI & Computer Vision | Open-Source Contributor

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PROFESSIONAL PROFILE

Full-Stack Engineer and ML/DL Developer with a mission to build products from 0 to production. Specialises in production-ready MERN applications, robust machine learning pipelines (data → model → API → dashboard), and high-performance computer vision systems. Internship-trained at Tata Advanced Systems Ltd. (TASL) — working on classified defence-grade simulation software — and VigyBag, a large open-source e-commerce platform. A self-driven engineer who manages the entire product lifecycle solo: React/Next.js interfaces to scalable Node.js/Python backends, ML model development, and cloud deployments on AWS with Docker and CI/CD.

Multi-hackathon winner (2x), peer-reviewed published researcher (JAAFR Journal 2025), and active open-source contributor recognised by GirlScript Summer of Code (GSSOC 2024). Enthusiastic about AI, computer vision, and building scalable systems that create real-world social impact.

TECHNICAL PROFICIENCIES

Languages & Frameworks: JavaScript (ES6+), TypeScript, Python, React.js, Next.js, Node.js, Express.js, FastAPI

Mobile & UI: React Native, Expo, Tailwind CSS, Shadcn/UI, Framer Motion, Bootstrap, Redux

Backend & Auth: REST APIs, WebSockets, WebRTC, BetterAuth, JWT, OAuth 2.0, Prisma, Drizzle, Mongoose

ML / AI / Computer Vision: TensorFlow, OpenCV, XGBoost, RAG, Groq AI, Deep Learning, Predictive Modeling, MLOps

Databases & Platforms: MongoDB, PostgreSQL, Redis, Convex, Supabase

Cloud & DevOps: AWS, Docker, CI/CD, Kubernetes, Nginx, Linux, Vercel, Git, GitHub

Tools & Methodologies: Postman, Jira, Bitbucket, Confluence, Agile, Scrum, Jest, Vite

WORK EXPERIENCE

VigyBag - Open-Source Software Engineer Intern

2024

Remote, India

- Contributed as an active developer on **VigyBag**, an open-source e-commerce platform bridging rural artisans with urban consumers across India (182+ GitHub stars, 419+ forks, 168+ contributors).
- Built and shipped React.js UI components and Node.js backend features for location-based product discovery and marketplace functionality.
- Worked on product search, category filtering, and pricing comparison across 10+ product categories, improving the vendor and buyer experience.
- Collaborated with a large open-source community; raised and resolved GitHub Issues, reviewed PRs, followed CI/CD and contribution guidelines.
- Stack: React.js, Redux, Node.js, Tailwind CSS, Vite, MongoDB, JavaScript, Vercel.

Tata Advanced Systems Ltd. (TASL) - Software Engineer Intern

2024 – 2025

Mumbai, India

- Engineered **CTS-71**, a confidential high-fidelity naval simulation platform replicating real-world Combat Management Systems for the Indian Navy, built on the MERN stack with WebSockets.
- Wrote automated test suites using Jest to ensure reliability and performance of mission-critical simulation modules.
- Collaborated with senior engineers in an Agile environment using Jira, Bitbucket, and Confluence for sprint planning and delivery.
- Received official offer letter and internship completion certificate from TASL upon successful delivery of all assigned modules.

PROJECTS

Samadhan — AI Civic Issue Reporting & Resolution Platform

[Demo](#)

Stack: Next.js, TypeScript, Convex, React Native (Expo), Tailwind CSS

- Won 1st place at Internal Hackathon 2025 hosted by Vidyalankar Polytechnic with a judging panel assessing technical innovation and real-world impact.
- Designed and built a full AI-driven civic issue reporting and resolution ecosystem allowing citizens to submit, track, and escalate municipal complaints with real-time status updates.
- Implemented cross-platform support using React Native (Expo) for mobile and Next.js for web, sharing a single Convex backend for real-time data sync.
- Delivered complete system — from database schema design and API architecture to UI implementation — within a 24-hour hackathon window.

Smart Meeter — AI Enterprise Meeting Platform

[Demo](#)

Stack: React, FastAPI, TensorFlow, OpenCV, Groq AI, WebRTC, Tailwind CSS

- Architected and built a full-featured AI-powered enterprise meeting platform featuring custom peer-to-peer WebRTC video conferencing without reliance on third-party SDKs.
- Integrated real-time AI transcription and Groq AI for fast NLP-based meeting summarisation, automatic action-item extraction, and decision logging.
- Built TensorFlow and OpenCV vision layers for participant engagement analytics and real-time video quality monitoring.
- Implemented a Kanban-based project workflow management system tied directly to meeting outcomes, allowing teams to turn decisions into tasks in real time.

Finch AI — Intelligent Banking Recommendation Engine

[Demo](#)

Stack: Next.js, Flask, XGBoost, Groq AI, PostgreSQL, MLOps

- Built an end-to-end ML-driven recommendation system serving personalised financial products (loans, credit cards, savings plans) to 500+ active users.
- Trained and tuned XGBoost classification models on financial transaction datasets, outperforming baseline regression and random forest models by a significant margin.
- Established full MLOps pipelines for continuous model retraining, A/B testing, drift monitoring, and deployment using Flask APIs and PostgreSQL.
- Integrated Groq AI for natural language financial advisory responses to complement the model's recommendations.

Cognix — Enterprise AI Assessment & Analytics Mobile App

[Demo](#)

Stack: React Native, Express.js, MongoDB, RAG (Retrieval-Augmented Generation), AWS, MLOps

- Designed and developed an enterprise-grade AI Assessment and Analytics mobile application for clinical cognitive testing and real-time cognitive evaluation in healthcare and HR contexts.
- Implemented RAG pipelines to generate contextually relevant, clinically grounded assessment questions dynamically based on user history and evaluation goals.
- Deployed the backend and ML inference services on AWS with autoscaling, load balancing, and a CI/CD pipeline for zero-downtime updates.
- Built a real-time analytics dashboard providing administrators with per-user cognitive performance trends, score distributions, and risk flags.

CTS-71 — Naval Combat Training Simulation (Confidential)

Stack: MERN (MongoDB, Express, React, Node.js), WebSockets, Jest | Client: Tata Advanced Systems Ltd. (TASL)

- Contributed to CTS-71, a high-fidelity classified naval simulation platform designed to replicate real-world Combat Management System (CMS) operations used aboard Indian Navy vessels.
- Developed real-time data streaming architecture using WebSockets to synchronise simulation state across multiple operator consoles simultaneously.
- Wrote comprehensive Jest test suites covering unit and integration testing for mission-critical modules, achieving high code coverage on all delivered features.

EDUCATION

Vidyalankar Polytechnic | **Diploma in Computer Engineering** | 2023 – 2026

Mumbai, Maharashtra, India

PUBLICATIONS & RESEARCH

- **Research Paper — JAAFR Journal (2025):** Peer-reviewed research publication in the field of Artificial Intelligence and Computer Engineering. Published independently during Diploma programme.

CERTIFICATIONS

- TensorFlow for Deep Learning — Udemy | Aug 2025
- Complete AI & Machine Learning, Data Science BootCamp — Udemy | Aug 2025
- Use Generative AI for Software Development — IBM SkillsBuild (Student Ambassador Program) | Dec 2024
- Explore Text-to-Speech Using IBM Watsonx — IBM SkillsBuild (Student Ambassador Program) | Dec 2024
- DevOps Discover — Microsoft Student Ambassadors – Imagine Cup | 2024
- Introduction to Large Language Models — Google Cloud Skills Boost | Sep 2024 | Credential ID: 11440118
- Introduction to Responsible AI — Google Cloud Skills Boost | Sep 2024 | Credential ID: 11440181
- Contributor in GSSOC Open Source — GirlScript Summer of Code | Aug 2024
- Learn Advanced C++ Programming — Udemy | Sep 2024
- C++ in Programming in Hindi — Great Learning | Nov 2024
- Data Structures in C — Great Learning | Nov 2024
- Visual Graphics in C — Great Learning | Sep 2024
- Postman API Fundamentals Student Expert — Postman | 2024
- The Complete Python Bootcamp From Zero To Hero — Udemy | Nov 2023

ACHIEVEMENTS

- **Industrial Hackathon 2026 — Winner:** TATA | Built an enterprise-grade AI solution under competition conditions.
- **Internal Hackathon 2025 — Winner:** Vidyalankar Polytechnic | Built Samadhan, an AI civic platform.
- **Research Paper Published:** JAAFR Journal (2025) | Peer-reviewed AI and engineering research.
- **INNOVGENIUS IDEATHON Participant:** Tata Consultancy Services (2025) | National-level ideathon by TCS.
- **SPIT Hackathon 2025 Participant:** Sardar Patel Institute of Technology | Inter-college competitive hackathon.
- **Top Contributor — GSSOC 2024:** GirlScript Summer of Code | Recognised as a top open-source contributor.
- **Open-Source Contributor — VigyBag:** 168+ contributors, 419+ forks | Active merged contributor.

OPEN-SOURCE CONTRIBUTIONS

- **VigyBag (github.com/codervivek5/VigyBag):** Active merged contributor to the VigyBag open-source e-commerce platform (182+ stars, 419+ forks, 168+ contributors). Contributed React UI components and Node.js backend features.
- **GirlScript Summer of Code 2024 (GSSOC):** Recognised as a Top Contributor in GSSOC 2024, one of India's largest open-source programmes. Received certificate of appreciation from GirlScript Foundation.

AVAILABILITY & FREELANCE SERVICES

Open to Work: Yes | Freelance: Available | Location: Mumbai, India (Remote-friendly)

Services: Full-Stack Web Apps (2–4 weeks) · ML Model + API (1–3 weeks) · Computer Vision Systems (2–3 weeks) · Cloud & CI/CD Setup (3–7 days) · Backend API Development (1–2 weeks) · Code Review (1–2 days)